

New metrics of affordable nutrition: which vegetables provide most nutrients for least cost?

Measuring food prices per gram, rather than per calorie, is one way to make healthful vegetables appear less expensive. However, a better measure of affordability would take the nutrient content of vegetables into account. This study, based on analyses of US Department of Agriculture datasets, aimed to identify which vegetables, including juices and soups, provided the most nutrients per unit cost. Nutrient density was measured using the Nutrient Rich Foods (NRF) index, based on nine nutrients to encourage: protein; fiber; vitamins A, C, and E; calcium; iron; magnesium; and potassium; and on three nutrients to limit: saturated fat, added sugar, and sodium. Food cost in dollars was calculated per 100 g, per 100 kcal, per serving, and per nutrient content. One-way analyses of variance with post hoc tests were used to determine statistical significance. Results showed that tomato juices and tomato soups, dark green leafy and nonleafy vegetables, and deep yellow vegetables, including sweet potatoes, had the highest NRF scores overall. Highest NRF scores per dollar were obtained for sweet potatoes, white potatoes, tomato juices and tomato soups, carrots, and broccoli. Tomato sauces, raw tomatoes, and potato chips were eaten more frequently than were many other vegetables that were both more affordable and more nutrient-rich. These new measures of affordable nutrition can help foodservice and health professionals identify those vegetables that provide the highest nutrient density per unit cost. Processed vegetables, including soups and juices, can contribute to the quality and the affordability of the diet. Copyright © 2013 Academy of Nutrition and Dietetics. Published by Elsevier Inc. All rights reserved.